Lab 6 Part 2 rubric of Truth, Justice, and Love!!!

No.	Topic	Points
2.1	**A Better BPF**	
	a. Generate Hamming filter. Plot frequency response. Measure/summarize as it says in Lab doc.	6
	b. Determine passband width. Make/Plot other BPFs. Measure/explain passband width as And how it changes with *L*. (Explanation worth most points)	6
	c. Determine (by hand) formula. Comment (i.e., explain) amplitudes of three components.	6
	d. Use frequency response to explain what it can pass.	2
2.2	**Piano Note Decoding**	
	(No points – this section is just an outline of what you will be doing!)	
2.3	**Piano Octaves**	
	a. Determine the upper/lower frequency edges for each octave (give in Hz and ω-ha	1
	b. Compute the center frequencies of the BPFs as the midpoint between the lower and upper edges	1
	NOTE: Please compile the above answers into an easily-readable table!	
2.4	**Bandpass Filter Bank Design**	
	a. Write code that determines β IN GENERAL (i.e., you need 1-2 lines that will work for all filters)	3
	b. Determine the lengths *L* for each filter to get the correct bandwidth. Show code! (Comment on your trial and error process for maximum partial credit)	6
	c. Plot the magnitude of the frequency responses all on one graph. Indicate the center frequencies and illustrate that the filters cover separate octaves.	1
	d. Zoom in to each filter and explain what you see. Answer question about passband/stopband.	5
2.5	**Piano Octave Decoding**	
	a. Generate the signal xx as shown	0
	b. Filter xx through the five filters from the previous part. Plot outputs in one figure.	1
	c. Validate the output signals (using the frequency response). Determine which octave the signals Are in.	2
	d. Answer question about how long transient lasts/whether it's different in each filter.	5
2.6	**Scoring for Activity**	
	a. Complete the octavescore function.	3
	b. Use all scores together and implement an automatic detection system.	7
	c. After filtering the xx signal through your system, comment and explain on how well your system Is working (the more correct explanation, the better!)	3
2.7	**Testing**	
	a. Test your system with our given test signal. Confirm your system works by comparing it t The “notes” vector.	6
	b. Do this test and comment on it well and I might consider giving a bit of extra credit!	0
	c. Do this test and comment on it well and I might consider giving a bit of extra credit!	0
	d. Comment on how well your system works. (Note; better systems may require less explanation, where systems with lots of errors will have a lot of explainin' to do~!)	6
	TOTAL:	70